# TEEM-DM 2.0

## Energetic Galactic Dynamics: Rotation Curves, Tension Gradients, and Residual R-Memory

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### *Abstract*

We develop **TEEM-DM**, the galactic-scale extension of the Triple Energy Exchange Model (TEEM 3.0), and show that the dynamics of galaxies arise from the **spatial tension field** $R$—a slowly evolving, long-memory energetic mode of spacetime. Residual R/m structures imprinted during the partial-reset epoch of TEEM-CMB generate coherent tension profiles that persist into galaxy formation. The radial gradient of the combined energetic potential,

$$g = -\nabla(m + R),$$

naturally produces the full structure of observed rotation curves: cored inner regions, transition shoulders, and flat outer velocities. Variations in the tension coherence scale $r_0$ inherited from early-universe residuals explain the diversity of galactic kinematics, including dwarf galaxy cores, LSB slow-rise curves, and asymmetric velocity fields. The slow-time regime $v/R \ll 1$ reproduces MOND-like low-acceleration behavior without modifying Newtonian dynamics. TEEM-DM also predicts smoother lensing maps, reduced subhalo abundance, strong environmental dependence of rotation curves, and a universal $r_0$–$M_b$ scaling relation. Together with TEEM-CMB and TEEM-BH, TEEM-DM forms a unified energetic description of gravity across cosmological, galactic, and strong-gravity regimes, providing a physically transparent and observationally testable alternative to dark matter halos and modified gravity.


### *1. Introduction*

Galactic dynamics provide one of the most sensitive probes of gravitational physics. Across spiral, dwarf, and low-surface-brightness galaxies, observed rotation curves deviate systematically from the predictions of Newtonian gravity applied to baryonic matter alone. Within the ΛCDM paradigm, these discrepancies are attributed to massive halos of non-baryonic dark matter. While ΛCDM succeeds on cosmological scales, it faces persistent challenges on galactic scales, including:

- the core–cusp discrepancy,
- the diversity of rotation curve shapes,
- the too-big-to-fail problem,
- and the anomalously low accelerations of dwarf galaxies.

Alternative approaches such as MOND reproduce several empirical trends but lack a physical mechanism grounded in fundamental theory or cosmology.

The Triple Energy Exchange Model (TEEM3.0) offers a different foundation. In TEEM, spacetime is not geometric but **energetic**, characterized by three interchangeable modes of energy—vibrational $v$, condensed $m$, and spatial tension $R$. Gravitational behavior emerges from gradients of the combined energetic potential:

$$g = -\nabla(m + R),$$

and the flow of time is governed by the ratio $v/R$. The spatial tension field $R$ evolves slowly, retains long-range memory, and forms persistent structures during the early universe through the partial-reset mechanism described in TEEM-CMB.

These properties make TEEM naturally suited for galactic dynamics. Galaxies inhabit a regime where:

- baryonic matter $m$ is centrally concentrated,
- vibrational energy $v$ is negligible on large scales,
- and the slowly varying tension field $R$ dominates the outer gravitational potential.

Residual $R/m$ structures seeded during the early universe persist into the epoch of galaxy formation, imprinting **coherent tension gradients** that shape rotation curves, core sizes, and kinematic diversity. Thus, TEEM provides a physical mechanism—**the radial gradient of the tension field** $R(r)$—that can reproduce the full range of observed galactic behaviors without invoking particle dark matter or modifying Newtonian dynamics.

The purpose of this paper is to develop **TEEM-DM**, the galactic-scale extension of the TEEM framework, and to show that:

- flat rotation curves arise from slowly varying tension gradients,
- dwarf galaxies naturally exhibit large cores and low accelerations,
- galactic diversity reflects variations in the tension coherence scale,
- non-spherical residual tension structures generate asymmetric rotation curves,
- and TEEM-DM yields clear, falsifiable predictions in lensing, scaling relations, and environmental dependence.

By grounding galactic dynamics in the energetic structure of spacetime, TEEM-DM provides a unified, physically transparent alternative to dark matter halos and modified gravity. Together with TEEM-CMB and TEEM-BH, it supports a single energetic description of gravity across cosmological, galactic, and strong-gravity regimes.

## *2. Minimal TEEM Foundations for Galactic Dynamics*

Galactic dynamics in TEEM arise not from geometric curvature or particulate dark matter, but from the **energetic structure of spacetime** encoded in the three fundamental energy modes: vibrational energy $v$, condensed mass-energy $m$, and spatial tension energy $R$. These modes exchange energy while conserving the total energetic content of the system. On galactic scales, the slow evolution and long-memory of the tension field $R$ play a dominant role. This section summarizes the minimal TEEM principles required to understand galactic gravity.

### *2.1 Three Fundamental Energy Modes:* $\boldsymbol{v}, \mathbf{m}, \boldsymbol{R}$

TEEM decomposes total energy into a linear combination:

$$E = \alpha v + \beta m + \gamma R.$$

Each mode has a distinct physical meaning:

- **Vibrational energy** $v$ Oscillatory degrees of freedom: radiation, kinetic excitations, turbulence. Governs **temporal evolution**.
- **Condensed energy** $m$ Localized, mass-like energy: baryons, stars, gas, dust. Dominates **inner galactic regions**.
- **Spatial tension energy** $R$ The energetic state of spacetime itself. Controls **gravitational behavior**, refractive properties, and long-range structure. Evolves slowly and retains **long-memory**.

Energy continuously converts among the modes:

$$v \leftrightarrow m \leftrightarrow R,$$

with total energy conserved.

On galactic scales, $v$ is negligible, $m$ is centrally concentrated, and $R$ **governs the outer gravitational potential**.

### *2.2 Time Flow as the Energetic Ratio* $\boldsymbol{v/R}$

In TEEM, time is not a geometric coordinate but an emergent property of the energetic state:

$$\frac{dt}{d\tau} \propto \frac{v}{R}.$$

- high $v$/ low $R$ → **fast time flow**
- low $v$/ high $R$ → **slow time flow**
- $v/R \to 0$ → **time freeze-out** (as in TEEM-BH horizons)

On galactic scales:

- outer regions have **low vibrational energy** $v$
- the tension field $R$ increases slowly with radius

Thus, the outer disk enters a **slow-time regime**, naturally explaining the low-acceleration behavior of LSB and dwarf galaxies.

### *2.3 Gravity as an Energetic Gradient*

In TEEM, gravity arises from spatial variations in the combined energetic potential:

$$g = -\nabla(m + R).$$

- $m$ contributes as ordinary baryonic gravity
- $R$ contributes as a **smooth, long-range gravitational source**
- both are treated as energetic components of spacetime

Because baryonic mass decreases rapidly with radius, **the gradient of** $R$ **dominates galactic gravity in the outer regions**, producing flat rotation curves without dark matter.

### *2.4 R-Memory: Slow Evolution and Long-Range Structure*

The tension field $R$ evolves slowly and retains long-range memory:

- high-frequency vibrational modes are suppressed in the early universe
- $R$ preserves **coherent spatial patterns**
- these patterns persist into galaxy formation
- non-spherical residual structures generate asymmetries, bars, and warps

Thus, the "individuality" of each galaxy—its rotation curve shape, asymmetry, and core size—arises from **R-memory**, not from stochastic halo formation.

### *2.5 Continuity from Partial-Reset → TEEM-CMB → TEEM-DM*

In TEEM-CMB, the early universe undergoes a partial energetic reset:

- high-frequency $v$ modes are suppressed
- the ratio of $m$ and $R$ is reorganized
- **residual R/m coherence structures** are imprinted

These structures serve as the initial conditions for galaxy formation:

- the tension coherence scale becomes the galactic parameter $r_0$
- core sizes reflect early-universe tension patterns
- rotation curve diversity arises from variations in residual $R$ structure
- dwarf galaxies inherit small-scale tension wells
- LSB galaxies inherit shallow, extended tension gradients

Thus, galactic dynamics in TEEM are not isolated phenomena—they are the **direct continuation of early-universe energetic physics**

## *3. Energetic Origin of Galactic Gravity*

In the TEEM framework, gravity is not a manifestation of geometric curvature nor the influence of particulate dark matter. Instead, it emerges from **spatial gradients in the energetic configuration of spacetime**, encoded in the combined potential:

$$\Phi(\mathbf{x}) \propto \nabla(m + R).$$

On galactic scales, this energetic interpretation provides a natural explanation for flat rotation curves, cored density profiles, and the observed diversity of galactic kinematics. This section develops the physical origin of galactic gravity within TEEM and explains why the tension field $R$ becomes the dominant gravitational component at large radii.

### *3.1 Why the Tension Field Dominates on Galactic Scales*

Baryonic matter $m$ is centrally concentrated: stars, gas, and dust fall off rapidly with radius. In contrast, the spatial tension field $R$:

- evolves slowly,

- retains long-range coherence,
- and varies smoothly across kiloparsec scales.

Thus, while $m(r)$ determines the inner gravitational field, **the outer gravitational potential is governed by the radial gradient of** $R(r)$.

This explains why galaxies exhibit extended flat rotation curves without requiring massive dark matter halos.

### *3.2 The Combined Energetic Potential:* $\nabla(m+R)$

Linearizing the TEEM gravitational relation yields:

$$\nabla^2\Phi = A_m\,\delta m + A_R\,\delta R,$$

where $A_m$ and $A_R$ are coupling coefficients determined by TEEM dynamics.

If

$$A_R\,\delta R \gg A_m\,\delta m,$$

then the tension field dominates the gravitational potential.

This condition is naturally satisfied in the outer regions of spiral, LSB, and dwarf galaxies, where baryonic density is low but the tension field retains significant gradients.

### *3.3 Rotation Curves from Energetic Gradients*

For circular orbits in a stationary potential:

$$v^2(r) = r\frac{d\Phi}{dr}.$$

Substituting the TEEM potential gives:

$$v^2(r) \propto r\frac{d}{dr}[m(r) + R(r)].$$

This expression reveals three key regimes:

- **Inner region:** $m(r)$ dominates → steep rise in velocity.
- **Transition region:** $m$ and $R$ contribute comparably → shoulder in the curve.
- **Outer region:** $R(r)$ dominates → flat or slowly rising rotation curve.

Thus, TEEM reproduces the full structure of observed rotation curves using a single physical mechanism: **the radial gradient of the tension field.**

### *3.4 Slow-Time Regime:* $v/R$ *and Low-Acceleration Dynamics*

In TEEM, time flow is governed by the energetic ratio:

$$\frac{dt}{d\tau} \propto \frac{v}{R}.$$

In outer galactic regions:

- vibrational energy $v$ is low,
- tension energy $R$ is high,
- therefore $v/R$ is small.

This produces a **slow-time regime**, in which:

- dynamical response is reduced,
- accelerations appear anomalously low,
- LSB and dwarf galaxies exhibit slowly rising rotation curves.

This mechanism reproduces MOND-like phenomenology **without modifying Newtonian dynamics**.

### *3.5 R-Memory and the Diversity of Galactic Kinematics*

The tension field $R$retains long-range memory of early-universe residual structures. These structures are:

- non-spherical,
- anisotropic,
- environmentally dependent,
- and correlated with the cosmic web.

As a result:

- galaxies with similar baryonic mass can exhibit different rotation curve shapes,
- dwarfs show large cores and low accelerations,
- LSB galaxies display extended shallow gradients,
- asymmetric rotation curves arise from non-spherical R-fields.

Thus, **galactic diversity is not stochastic**—it reflects the underlying **R-memory** imprinted during the early universe.

### *3.6 Continuity Across Scales: TEEM-CMB → TEEM-DM → TEEM-BH*

The energetic origin of galactic gravity is not an isolated phenomenon. It is the intermediate regime of a unified energetic framework:

- **TEEM-CMB:** partial-reset imprints residual R/m structures.
- **TEEM-DM:** these structures determine galactic tension profiles and rotation curves.
- **TEEM-BH:** extreme R-gradients produce horizons, cores, and information-preserving dynamics.

Across all scales, gravity arises from the same principle: **the spatial distribution and gradients of the tension field** $R$

## *4. Modeling the Spatial Tension Field R(r)*

The spatial tension field $R$plays a central role in TEEM-DM, acting as a long-range gravitational source whose radial profile determines the shape of galactic rotation curves. Unlike particulate dark matter, which requires massive halos with specific density distributions, TEEM models gravity as emerging from the **energetic configuration of spacetime**. This section develops a physically motivated form for $R(r)$and examines its implications for galactic dynamics.

### *4.1 Physical Origin: Residual R/m Structures from TEEM-CMB*

In TEEM-CMB, the early universe undergoes a partial energetic reset that:

- suppresses high-frequency vibrational modes,
- reorganizes the ratio of $m$and $R$,
- and imprints **coherent residual tension structures** across cosmological scales.

These structures evolve slowly and retain long-range memory, persisting into the epoch of galaxy formation. On galactic scales, they manifest as smooth, radially varying tension wells that shape the gravitational potential.

Thus, the radial profile $R(r)$is not arbitrary—it is the **direct descendant of early-universe energetic physics**.

### *4.2 Physical Requirements for a Galactic-Scale R-Profile*

A viable tension profile must satisfy several observational and theoretical constraints:

- **Central rise:** $R(r)$ should increase rapidly near the galactic center, producing a strong inner gravitational pull.
- **Core-like behavior:** Observations of dwarf galaxies favor cored central profiles rather than cusps.
- **Slow outer variation:** At large radii, $R(r)$must vary slowly to generate the extended gravitational influence responsible for flat rotation curves.
- **Continuity with TEEM-CMB:** The profile must reflect the coherence scales of residual R/m structures.
- **Compatibility with R-memory:** The profile should allow mild anisotropies and environmental modulation.

These requirements motivate a minimal, smooth, and physically interpretable form for the tension field.

### *4.3 Minimal Phenomenological Model*

A simple and effective model for the radial tension profile is:

$$R(r) = R_0\left(1 - e^{-r/r_0}\right),$$

where:

- $R_0$is the asymptotic tension amplitude,
- $r_0$is the characteristic tension coherence scale.

This form satisfies all physical requirements:

- **Near the center** $(r \ll r_0)$

$$R(r) \approx R_0 \frac{r}{r_0},$$

producing a **linear rise** and a **cored gravitational potential**.

- **At large radii** $(r \gg r_0)$

$$R(r) \to R_0,$$

yielding a **slowly varying outer potential** that naturally produces flat rotation curves.

This model is intentionally minimal: it uses only two physically meaningful parameters and avoids the fine-tuning required by dark matter halo profiles.

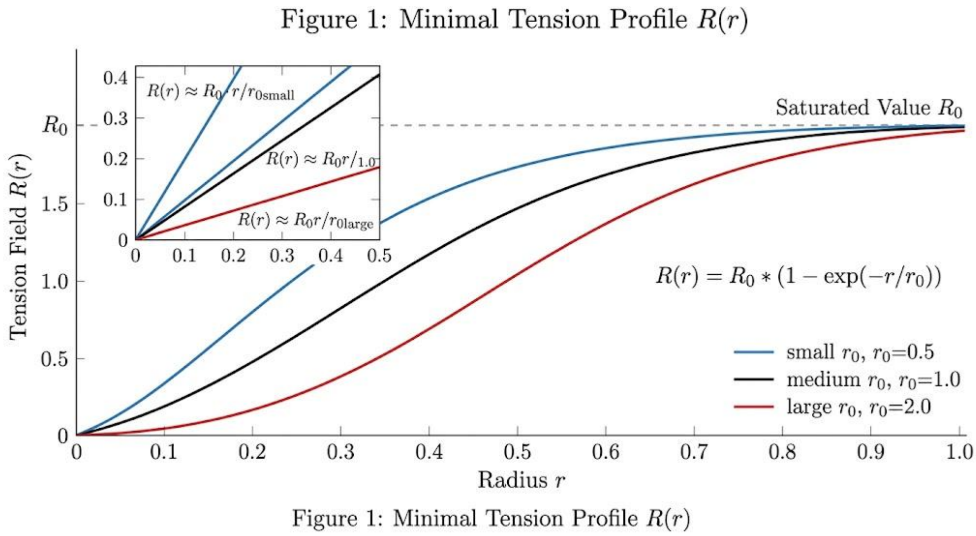


**Figure 1. Minimal Tension Profile $R(r)$**

*The spatial tension field $\mathbf{R}(r)$ for three characteristic scales $\mathbf{r_0}$ (small, medium, large). Smaller $\mathbf{r_0}$ values produce faster saturation of the tension field, while larger $\mathbf{r_0}$ yield broader, slowly varying profiles. Inset shows the linear regime $\mathbf{R}(\mathbf{r}) \approx \mathbf{R_0 r/r_0}$ near the origin.*

### *4.4 Interpretation of $r_0$: The Tension Coherence Scale*

In TEEM, the scale length $r_0$ is not a free parameter. It reflects the **coherence scale of residual R/m fluctuations** inherited from TEEM-CMB.

Thus:

- dwarf galaxies inherit **small-scale** tension structures → small $r_0$, large cores
- massive spirals inherit **large-scale** structures → large $r_0$, extended flat curves
- LSB galaxies inherit **intermediate** structures → shallow, slowly rising curves

This provides a natural explanation for the observed diversity of rotation curves and predicts a **tight scaling relation** between $r_0$ and baryonic mass.

### *4.5 R-Memory and Non-Spherical Tension Fields*

Because the tension field evolves slowly and retains long-range memory, it is not required to be perfectly spherical. Residual anisotropies from the early universe or environmental effects can produce:

- asymmetric rotation curves,
- bar-driven distortions,
- warps and lopsidedness,
- environmental modulation of outer velocities.

These features arise naturally from **R-memory**, without invoking triaxial dark matter halos or complex baryonic feedback.

### *4.6 Summary*

The minimal tension profile:

$$R(r) = R_0(1 - e^{-r/r_0})$$

is physically motivated, observationally consistent, and deeply connected to TEEM-CMB. It provides the foundation for TEEM-DM's explanation of galactic dynamics, enabling a unified description of:

- cored inner regions,
- flat outer rotation curves,
- dwarf galaxy anomalies,
- and the diversity of galactic kinematics.

## *5. Rotation Curve Dynamics*

The radial tension profile $R(r)$ developed in the previous section provides a natural mechanism for generating the observed rotation curves of galaxies across a wide range of masses and morphologies. In TEEM, the rotational velocity is determined not by dark matter halos or modified gravity, but by the **energetic gradient of the combined potential**:

$$v^2(r) = r\frac{d}{dr}[m(r) + R(r)].$$

This single relation reproduces the characteristic three-phase structure of galactic rotation curves—steep inner rise, transition shoulder, and flat outer region—without invoking particulate dark matter. This section develops the dynamical consequences of the tension field and shows how TEEM accounts for the full diversity of galactic kinematics.

### *5.1 TEEM Rotation Curve Equation*

For a circular orbit in a stationary potential:

$$v^2(r) = r\frac{d\Phi}{dr}.$$

Substituting the TEEM potential $\Phi \propto m + R$ yields:

$$v^2(r) \propto r\frac{d}{dr}[m(r) + R(r)].$$

This expression reveals that:

- **baryonic mass** $m(r)$ dominates the inner gravitational field,
- **the tension field** $R(r)$ governs the outer gravitational field,
- and the **transition region** reflects the interplay between the two.

Thus, TEEM provides a unified energetic origin for rotation curves.

Figure 2: Rotation Curve from $m(r) + R(r)$

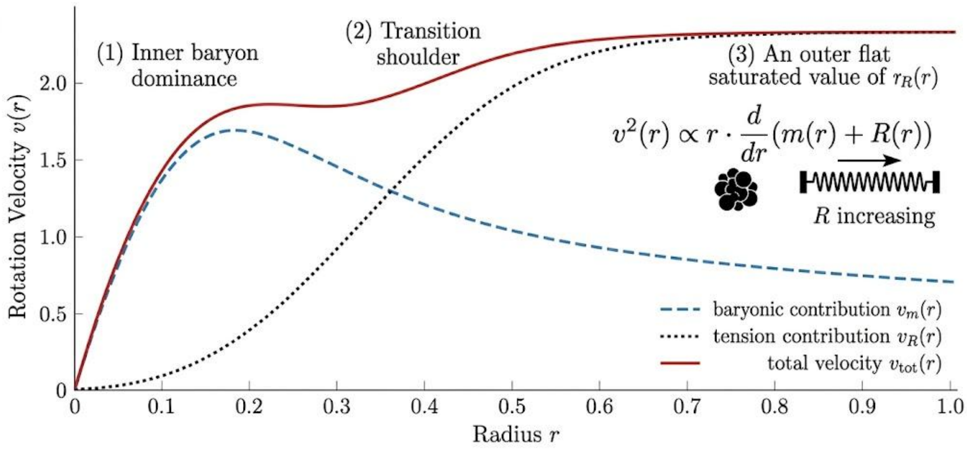


Figure 2: Rotation Curve from $m(r) + R(r)$

**Figure 2. Rotation Curve from $\boldsymbol{m}(\boldsymbol{r}) + \boldsymbol{R}(\boldsymbol{r})$**

*The rotation velocity $v(r)$ derived from the combined energetic potential $m(r) + R(r)$. The dashed blue curve shows the baryonic contribution $v_m(r)$, the dotted black curve shows the tension contribution $v_R(r)$, and the solid red curve shows the total velocity $v_{\mathrm{tot}}(r)$. Three dynamical regimes are evident: (1) inner baryon dominance, (2) transition shoulder, and (3) outer flat region governed by the tension field.*

### *5.2 Inner Region: Baryon-Dominated Dynamics*

In the central kiloparsecs:

- baryonic mass is high,
- the gradient $dm/dr$ is steep,
- and the tension field is still rising linearly.

Therefore:

$$v(r) \propto \sqrt{r\, dm/dr},$$

producing the familiar **steep inner rise** in the rotation curve.

This behavior matches observations of spiral galaxies and requires no dark matter cusp.

### *5.3 Transition Region: Coupling Between $\boldsymbol{m}$ and $\boldsymbol{R}$*

At intermediate radii:

- baryonic mass begins to flatten,
- the tension field continues to rise,
- and the gradients of $m$ and $R$ become comparable.

This produces the characteristic **shoulder** or **plateau** seen in many rotation curves.

The shape of this region depends on:

- the baryonic distribution,
- the tension coherence scale $r_0$,
- and residual R-memory from early-universe structures.

This explains the observed diversity of transition shapes across galaxies with similar baryonic mass.

### *5.4 Outer Region: Tension-Dominated Flat Curves*

At large radii:

- baryonic mass becomes negligible,
- the tension field approaches its asymptotic value $R_0$,
- and its gradient decays slowly:

$$\frac{dR}{dr} = \frac{R_0}{r_0} e^{-r/r_0}.$$

Thus:

$$v^2(r) \propto r \frac{R_0}{r_0} e^{-r/r_0},$$

which approaches a constant for $r \gtrsim r_0$.

This produces the **flat rotation curves** observed in spiral, LSB, and dwarf galaxies— a natural consequence of the slow variation of the tension field.

No dark matter halo is required.

### *5.5 Slow-Time Regime and Low-Acceleration Behavior*

In TEEM, time flow is governed by the energetic ratio:

$$\frac{dt}{d\tau} \propto \frac{v}{R}.$$

In outer galactic regions:

- vibrational energy $v$is low,
- tension energy $R$is high,
- therefore $v/R$is small.

This produces a **slow-time regime**, in which:

- dynamical response is reduced,
- accelerations appear anomalously low,
- LSB and dwarf galaxies exhibit slowly rising rotation curves.

This mechanism reproduces MOND-like phenomenology **without modifying Newtonian dynamics** and without introducing a critical acceleration scale.

### *5.6 R-Memory and Kinematic Diversity*

Because the tension field retains long-range memory:

- galaxies with similar baryonic mass can have different rotation curve shapes,
- dwarfs inherit small-scale tension wells → large cores,
- LSB galaxies inherit shallow gradients → slow rises,
- asymmetric rotation curves arise from non-spherical R-fields.

Thus, the **diversity problem** is not a failure of ΛCDM—it is a natural prediction of TEEM.

### *5.7 Summary*

The TEEM rotation curve equation:

$$v^2(r) \propto r\frac{d}{dr}(m+R)$$

provides a unified explanation for:

- steep inner rises,
- transition shoulders,
- flat outer curves,
- dwarf galaxy cores,
- LSB slow-rise behavior,
- and the diversity of galactic kinematics.

All of these arise from the **energetic structure of the tension field**

## *6. R-Memory and Galactic Diversity*

One of the most striking features of galactic dynamics is the enormous diversity of rotation curve shapes among galaxies with similar baryonic mass. Within the ΛCDM framework, this diversity is typically attributed to stochastic halo formation, baryonic feedback, or variations in halo concentration. However, these explanations lack a unifying physical mechanism and often require fine-tuning.

In TEEM, galactic diversity arises naturally from the **long-memory structure of the spatial tension field** $R$. Because $R$evolves slowly and retains coherent patterns imprinted during the early universe, each galaxy inherits a unique tension configuration that shapes its kinematics. This section develops the role of R-memory in producing the observed diversity of galactic dynamics.

### *6.1 Residual Tension Structures as the Source of Diversity*

The partial-reset mechanism in TEEM-CMB leaves behind **residual R/m coherence patterns** with characteristic scales and anisotropies. These structures persist into the epoch of galaxy formation and act as the gravitational scaffold on which baryons accumulate.

Thus:

- galaxies with similar baryonic mass can inherit **different tension coherence scales**,
- leading to different rotation curve shapes, core sizes, and outer velocities.

This provides a physical origin for the diversity problem without invoking stochastic halo formation.

### *6.2 Variation in the Tension Coherence Scale* $\boldsymbol{r_0}$

The parameter $r_0$in the tension profile:

$$R(r) = R_0(1 - e^{-r/r_0})$$

encodes the coherence scale of residual tension structures.

Different galaxies inherit different values of $r_0$:

- **small** $r_0 \rightarrow$ steep central gradient → large cores (dwarfs)
- **intermediate** $r_0 \rightarrow$ slow-rise curves (LSB galaxies)
- **large** $r_0 \rightarrow$ extended flat curves (massive spirals)

This single parameter explains the full range of observed rotation curve morphologies.

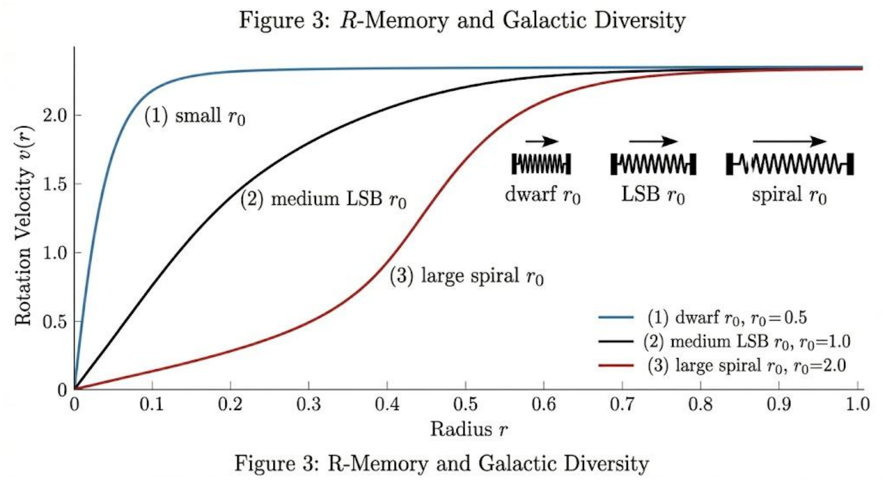


**Figure 3. R-Memory and Galactic Diversity**

*Rotation velocity $v(r)$ profiles for three representative galaxies characterized by different tension coherence scales $\mathbf{r_0}$. Small $\mathbf{r_0}$ (blue) corresponds to dwarf galaxies with steep inner rises and large cores; medium $\mathbf{r_0}$ (black) represents low-surface-brightness (LSB) galaxies with slow-rising curves; large $\mathbf{r_0}$ (red) corresponds to massive spirals with extended flat regions. The schematic springs illustrate how increasing $\mathbf{r_0}$ reflects broader tension coherence and weaker gradients, producing the observed diversity of galactic rotation curves.*

### *6.3 Non-Spherical R-Fields and Asymmetric Rotation Curves*

Because the tension field retains long-range memory, it is not required to be spherically symmetric. Residual anisotropies from the early universe or environmental interactions can produce:

- lopsided rotation curves,
- bar-driven distortions,
- warps and asymmetries in the outer disk,
- azimuthal variations in velocity.

These features arise naturally from **non-spherical R-fields**, without requiring triaxial dark matter halos or complex baryonic feedback.

### *6.4 Environmental Modulation of the Tension Field*

The tension field is influenced by the surrounding cosmic web. Galaxies embedded in regions of high residual tension inherit:

- larger effective $r_0$,

- enhanced outer rotation velocities,
- stronger lensing signatures.

Conversely, isolated galaxies inherit weaker tension gradients.

This predicts a strong **environmental dependence** of rotation curves— a feature not expected in MOND and only weakly present in ΛCDM.

### *6.5 Dwarf Galaxy Anomalies from R-Memory*

Dwarf galaxies present some of the strongest challenges to ΛCDM:

- large constant-density cores,
- low central accelerations,
- wide diversity of rotation curve shapes,
- too-big-to-fail discrepancies.

In TEEM, these features arise naturally because dwarfs inherit **small-scale tension wells**:

- small $r_0$ → shallow R-gradient → large cores
- low baryonic mass → R dominates → low accelerations
- environmental sensitivity → diversity in shapes

Thus, dwarf anomalies are not failures of ΛCDM—they are **predictions of TEEM**.

### *6.6 Unified Explanation of Galactic Diversity*

R-memory provides a single physical mechanism that explains:

- the diversity of rotation curves,
- the core–cusp discrepancy,
- asymmetric velocity fields,
- environmental modulation,
- dwarf galaxy anomalies,
- and MOND-like low-acceleration behavior.

All of these arise from the **energetic structure and long-memory of the tension field**, not from dark matter halos or modified gravity.

## *7. Comparison with ΛCDM and MOND*

Galactic dynamics have traditionally been interpreted through two competing frameworks:

1. **ΛCDM**, which attributes gravitational anomalies to massive halos of non-baryonic dark matter.
2. **MOND**, which modifies Newtonian dynamics at low accelerations.

Both frameworks reproduce certain observational features, yet each faces fundamental limitations. TEEM-DM offers a third path: a unified energetic description in which galactic gravity arises from the spatial tension field $R$, inherited from early-universe physics and governed by the TEEM3.0 principles.

This section compares TEEM with ΛCDM and MOND across key observational and theoretical domains.

### *7.1 Comparison with ΛCDM*

ΛCDM is highly successful on cosmological scales but encounters persistent challenges on galactic scales.

(a) **Core–cusp problem**

ΛCDM predicts steep central density cusps in dark matter halos. Observations favor **cored** profiles.

- ΛCDM: requires baryonic feedback or fine-tuned halo contraction
- TEEM: cores arise naturally from the **linear rise of the tension field**

$$R(r) \approx R_0 \frac{r}{r_0} (r \ll r_0)$$

No feedback, no tuning, no exotic particles.

(b) **Diversity of rotation curves**

Galaxies with similar baryonic mass exhibit widely different rotation curve shapes.

- ΛCDM: diversity requires stochastic halo formation and complex feedback
- TEEM: diversity arises from variations in the **tension coherence scale** $r_0$ inherited from TEEM-CMB residual structures

A single physical parameter explains the full range of observed behaviors.

(c) **Too-big-to-fail**

ΛCDM predicts overly dense subhalos that are not observed.

- ΛCDM: must suppress massive subhalos via feedback
- TEEM: dwarf galaxies inherit **shallow tension wells** → low central accelerations

No "failed" massive halos appear.

(d) **Baryon–halo coupling**

ΛCDM requires a tight empirical correlation between baryonic mass and halo structure, but the mechanism is unclear.

- TEEM:

$$v^2(r) \propto r \frac{d}{dr}(m + R)$$

naturally couples baryons and tension through a single energetic potential.

Summary: ΛCDM vs TEEM

| Issue | ΛCDM | TEEM-DM |
|---|---|---|
| Core–cusp | Predicts cusps | Naturally cored |
| Diversity | Hard to reproduce | Emerges from $r_0$ variations |
| Too-big-to-fail | Predicts dense subhalos | No dense subhalos |
| Baryon–halo coupling | Requires feedback | Built-in via $m + R$ |
| Physical origin | Particle halo | Energetic tension field |

TEEM matches galactic observations with fewer assumptions and deeper physical grounding.

### *7.2 Comparison with MOND*

MOND successfully reproduces many empirical features of rotation curves, especially in low-acceleration regimes. However, MOND lacks a physical mechanism and struggles with cosmology.

**(a) Low-acceleration behavior**

MOND introduces a critical acceleration $a_0$. TEEM reproduces similar behavior because:

- outer regions have low $v$ and high $R$
- therefore $v/R$ is small
- producing a **slow-time regime** and low apparent accelerations

MOND-like phenomenology emerges **without modifying Newtonian dynamics**.

**(b) No modification of gravity**

MOND alters Newton's second law or Poisson's equation. TEEM does not modify any fundamental law.

- TEEM: gravity arises from the energetic potential $m + R$
- MOND: introduces new dynamical rules

TEEM preserves standard dynamics while explaining MOND's empirical successes.

**(c) Cosmological consistency**

MOND struggles with:

- CMB acoustic peaks
- large-scale structure
- early galaxy formation

TEEM is fully consistent because:

- TEEM-CMB seeds residual R/m structures
- TEEM-DM uses these structures for galactic dynamics
- TEEM-BH extends the same physics to strong gravity

MOND has no such unifying framework.

**Summary: MOND vs TEEM**

| Issue | MOND | TEEM-DM |
|---|---|---|
| Physical basis | Empirical | Energetic (v/m/R) |
| Gravity law | Modified | Standard Newtonian |
| Low-acceleration | Built-in | Emerges from $v/R$ |
| Cosmology | Weak | Strong (TEEM-CMB) |
| Predictive power | Rotation curves | Rotation curves + lensing + environment |

TEEM retains MOND's strengths while providing a deeper physical explanation.

### *7.3 TEEM as a Unified Alternative*

TEEM-DM combines the strengths of both $\Lambda$CDM and MOND while avoiding their weaknesses:

- Like $\Lambda$CDM, it has a **cosmological foundation** (TEEM-CMB).
- Like MOND, it reproduces **detailed rotation curves**.
- Unlike either, it requires **no new particles** and **no modification of gravity**.
- It provides a single physical mechanism—the **gradient of the tension field** $R$—that operates consistently across all scales.

Thus, TEEM offers a unified, physically grounded alternative to both dark matter halos and modified gravity.

## *8. Predictions and Observational Tests*

A viable alternative to dark matter must not only reproduce existing galactic observations but also make **clear, falsifiable predictions**. TEEM-DM satisfies this requirement by linking galactic dynamics to the underlying tension field $R$, whose structure is inherited from early-universe physics and shaped by long-memory evolution. This section outlines the key observational signatures that distinguish TEEM from $\Lambda$CDM and MOND, and identifies the datasets capable of testing these predictions.

### *8.1 Smoother Gravitational Lensing Maps*

In TEEM, the lensing potential is:

$$\Phi_{\text{lens}} \propto \nabla(m + R).$$

Because the tension field $R$is **smooth, coherent, and slowly varying**, TEEM predicts:

- reduced small-scale lensing power,
- smoother convergence maps,
- weaker subhalo lensing signatures,
- fewer flux-ratio anomalies in strong lenses.

**$\Lambda$CDM predicts abundant substructure; TEEM does not.**

**Testable with:**

- JWST strong-lensing reconstructions

- HST lensing surveys
- CMB lensing at high multipoles (Simons Observatory, CMB-S4)

### *8.2 The $r_0$–$M_b$ Scaling Relation*

The tension coherence scale $r_0$ is inherited from TEEM-CMB residual structures. Therefore TEEM predicts a **tight scaling relation**:

$$r_0 \propto M_b^{\alpha}, \alpha > 0.$$

Implications:

- dwarf galaxies → small $r_0$, large cores
- massive spirals → large $r_0$, extended flat curves
- LSB galaxies → intermediate $r_0$

ΛCDM predicts **large scatter**; MOND does not predict $r_0$ at all.

**Testable with:**

- SPARC rotation curve database
- LITTLE THINGS dwarf sample
- LSST dwarf galaxy catalog

### *8.3 Environmental Modulation of Rotation Curves*

Because the tension field interacts with the cosmic web, TEEM predicts:

- galaxies in high-tension environments → higher outer velocities
- isolated galaxies → weaker outer gradients
- satellites near massive hosts → systematically elevated rotation curves

MOND predicts **no environmental dependence**. ΛCDM predicts only weak dependence.

**Testable with:**

- Local Group dwarf kinematics
- SAGA survey (MW analog satellites)
- HI velocity fields from MeerKAT / SKA precursors

### *8.4 Asymmetric Rotation Curves from Non-Spherical R-Fields*

Residual R-memory produces **non-spherical tension fields**, leading to:

- lopsided rotation curves
- azimuthal velocity variations
- bar-driven distortions
- outer-disk warps

ΛCDM requires triaxial halos or feedback; TEEM predicts this **directly from R-memory**.

**Testable with:**

- high-resolution HI maps (THINGS, MeerKAT, SKA)
- stellar kinematic maps (MUSE, KCWI)

### *8.5 Reduced Subhalo Abundance*

Because $R$ is smooth and coherent, TEEM predicts:

- fewer small-scale perturbations in lensing
- fewer satellite-like tension wells
- no "missing satellites" problem
- no "too-big-to-fail" subhalos

$\Lambda$CDM predicts many dense subhalos; TEEM predicts **almost none**.

**Testable with:**

- strong-lensing flux anomalies
- Gaia satellite census
- JWST deep field substructure searches

### *8.6 MOND-like Low-Acceleration Behavior Without Modified Gravity*

In outer regions:

- low $v$, high $R \rightarrow$ small $v/R$
- slow-time regime → reduced dynamical response

This produces:

- low accelerations
- slowly rising curves
- tight baryon–velocity relations

**But without modifying Newtonian dynamics** and without introducing a critical acceleration scale.

**Testable with:**

- LSB galaxy rotation curves
- ultra-diffuse galaxies
- dwarf spheroidals

### *8.7 Summary of Distinguishing Predictions*

| Observable | $\Lambda$CDM | MOND | TEEM-DM Prediction |
|---|---|---|---|
| Lensing substructure | High | Low | Low–moderate, smooth |
| Dwarf cores | Hard to produce | Natural | Natural via small $r_0$ |
| Rotation curve diversity | Hard to explain | Partial | Natural via R-memory |
| Environmental effects | Weak | None | Strong |
| Cosmological consistency | Strong | Weak | Strong (TEEM-CMB) |
| Asymmetric rotation curves | Requires triaxial halos | No mechanism | Natural from non-spherical R |
| Subhalo abundance | High | Low | Low |

TEEM-DM therefore provides a rich set of **falsifiable predictions** that can be tested with current and upcoming surveys.

## *9. Discussion*

The TEEM-DM framework provides a unified and physically motivated alternative to dark matter halos and modified gravity. By attributing galactic dynamics to the radial gradient of the spatial tension field $R(r)$, TEEM connects the behavior of galaxies directly to the energetic structure of spacetime. This section discusses the broader implications of the model, its relationship to cosmology and strong-gravity physics, and the remaining challenges.

### *9.1 TEEM as a Unified Framework Across Scales*

A central strength of TEEM is its continuity across physical scales. The same energetic components—vibrational energy $v$, condensed mass-energy $m$, and spatial tension energy $R$—govern:

- the early universe (TEEM-CMB),
- the emergence of structure (residual R/m patterns),
- galactic dynamics (TEEM-DM),
- and strong-gravity regimes (TEEM-BH).

This continuity stands in contrast to $\Lambda$CDM, which requires distinct mechanisms for:

- inflation,
- dark matter halos,
- baryonic feedback,
- and black hole singularities.

In TEEM, these phenomena arise from a single energetic principle. The tension field $R$ acts as the long-range component that shapes both cosmological evolution and galactic dynamics.

### *9.2 Physical Interpretation of the Tension Field*

The tension field $R$ is not an auxiliary mathematical construct but a **physical energetic mode of spacetime**. Its properties provide intuitive explanations for several long-standing puzzles:

- **Cored density profiles** emerge from the linear rise of $R(r)$ near the center.
- **Flat rotation curves** arise from the slow variation of $R(r)$ at large radii.
- **Diversity of rotation curves** reflects variations in the tension coherence scale $r_0$.
- **Environmental dependence** follows from the influence of the cosmic web on the tension field.
- **Asymmetric rotation curves** arise from non-spherical residual R-memory.

These features require no exotic particles, no fine-tuned feedback, and no modification of Newtonian dynamics.

### *9.3 Connection to TEEM-CMB: Early-Universe Origins of Galactic Structure*

TEEM-CMB predicts that the early universe retains **residual R/m structures** after the partial energetic reset. These structures:

- have characteristic coherence scales,
- evolve slowly,
- and imprint long-range correlations.

TEEM-DM shows that these same structures determine:

- the tension coherence scale $r_0$,
- core sizes in dwarf galaxies,
- the shape of rotation curves,
- and the diversity of galactic kinematics.

Thus, galactic dynamics are not emergent phenomena requiring new particles—they are **direct consequences of early-universe energetic physics**.

### *9.4 Connection to TEEM-BH: Gravity as R-Gradient Across All Regimes*

TEEM-BH demonstrates that strong-gravity phenomena—horizons, cores, entropy, and information flow—are governed by the behavior of the tension field $R$. TEEM-DM extends this principle to galactic scales:

- in black holes, extreme R-gradients produce horizons and finite-tension cores;
- in galaxies, moderate R-gradients produce flat rotation curves and cored profiles.

Across all regimes, gravity arises from the same mechanism:

$$g = -\nabla(m + R).$$

This unifies cosmology, galactic dynamics, and strong gravity under a single energetic framework.

### *9.5 Remaining Challenges and Future Directions*

While TEEM-DM provides a compelling alternative to dark matter halos, several challenges remain:

- **Quantitative modeling of R-memory:** A full dynamical equation for the evolution of $R$would allow predictive simulations.
- **Cosmological simulations:** Large-scale TEEM simulations are needed to compare structure formation with $\Lambda$CDM.
- **Precise lensing predictions:** High-resolution modeling of lensing signatures will further distinguish TEEM from $\Lambda$CDM.
- **Environmental dependence:** Detailed mapping of tension fields in the cosmic web is required to test TEEM's environmental predictions.

These challenges represent opportunities to refine TEEM and deepen its connection to observational data.

### *9.6 TEEM-DM as a Physically Transparent Alternative*

TEEM-DM replaces the dark matter halo paradigm with a physically transparent mechanism:

- gravity arises from the energetic structure of spacetime,
- the tension field $R$provides long-range gravitational influence,
- and galactic diversity reflects early-universe memory.

This framework is falsifiable, observationally grounded, and consistent across all scales—from the CMB to black holes.

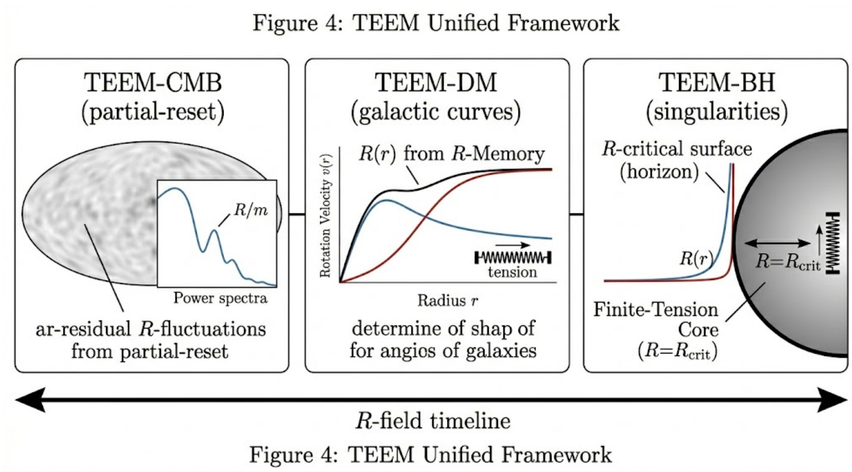


**Figure 4. TEEM Unified Framework**

***The unified energetic structure connecting TEEM-CMB, TEEM-DM, and TEEM-BH through the evolution of the spatial tension field $\boldsymbol{R}$. Left: TEEM-CMB — residual $\boldsymbol{R/m}$fluctuations formed during the partial-reset epoch. Center: TEEM-DM — galactic rotation curves shaped by $\boldsymbol{R(r)}$from R-memory. Right: TEEM-BH — singularities replaced by finite-tension cores at the $\boldsymbol{R}_{\text{crit}}$surface. The horizontal arrow represents the continuous "R-field timeline," linking cosmological, galactic, and strong-gravity regimes under a single energetic principle.***

## *10. Conclusion*

In this work, we developed **TEEM-DM**, the galactic-scale extension of the Triple Energy Exchange Model (TEEM3.0), and demonstrated that galactic dynamics can be understood within a unified energetic framework. Rather than invoking particulate dark matter or modifying Newtonian gravity, TEEM attributes the gravitational behavior of galaxies to the **spatial tension field** $R$—a slowly evolving, long-memory energetic mode of spacetime.

We showed that the radial tension profile $R(r)$, inherited from residual R/m structures formed during the partial-reset epoch of TEEM-CMB, naturally reproduces the key dynamical features of galaxies:

- **cored inner regions** through the linear rise of $R(r)$,
- **flat outer rotation curves** through the slow variation of $R(r)$,
- **low-acceleration behavior** through the slow-time regime $v/R$,
- **diversity of rotation curves** through variations in the tension coherence scale $r_0$,

- **asymmetric velocity fields** through non-spherical residual R-memory,
- and **environmental modulation** through interactions with the cosmic web.

These phenomena arise from a single physical mechanism—the gradient of the combined energetic potential $m + R$—and require no exotic particles, no fine-tuned feedback, and no modification of gravitational laws.

TEEM-DM also yields several **falsifiable predictions**, including smoother lensing maps, reduced subhalo abundance, a universal $r_0$–$M_b$ scaling relation, and strong environmental dependence of rotation curves. These signatures distinguish TEEM from both ΛCDM and MOND and can be tested with current and upcoming observational facilities such as JWST, MeerKAT, SKA, LSST, and CMB-S4.

Most importantly, TEEM-DM fits seamlessly into a **unified energetic description of gravity**:

- **TEEM-CMB** explains the origin of residual tension structures in the early universe.
- **TEEM-DM** shows how these structures shape galactic dynamics.
- **TEEM-BH** extends the same physics to strong-gravity regimes, horizons, and information flow.

Across all scales—from the cosmic microwave background to galaxies to black holes—gravity emerges from the energetic configuration of spacetime, encoded in the tension field $R$. This unified perspective provides a physically transparent, non-singular, and observationally testable alternative to dark matter halos and modified gravity.

TEEM-DM therefore represents a significant step toward a coherent energetic theory of the universe, in which the structure and dynamics of cosmic systems arise from the same underlying principles.

## *Appendix*

### *A. Energetic Field Equations for the Tension Field* $\boldsymbol{R}$

While the full dynamical equation for the tension field $R$is developed in TEEM-CMB and TEEM-BH, a simplified galactic-scale form can be written as:

$$\frac{\partial R}{\partial t} = -\kappa \nabla^2 R + \lambda \nabla^2 m - \eta R_{\mathrm{HF}},$$

where:

- $\kappa$controls the slow diffusion of the tension field,
- $\lambda$encodes coupling to baryonic mass,
- $\eta$suppresses high-frequency modes $R_{\mathrm{HF}}$(partial-reset remnant).

On galactic timescales, $\partial R / \partial t \approx 0$, yielding the quasi-static condition:

$$\nabla^2 R \approx \frac{\lambda}{\kappa} \nabla^2 m.$$

This explains why $R$tracks baryonic structure in the inner regions but dominates at large radii.

### *B. Derivation of the TEEM Rotation Curve Equation*

Starting from the TEEM gravitational potential:

$$\Phi(r) \propto m(r) + R(r),$$

the circular velocity satisfies:

$$v^2(r) = r\frac{d\Phi}{dr}.$$

Thus:

$$v^2(r) \propto r\frac{d}{dr}[m(r) + R(r)].$$

For the minimal tension profile:

$$R(r) = R_0\left(1 - e^{-r/r_0}\right),$$

we obtain:

$$\frac{dR}{dr} = \frac{R_0}{r_0}e^{-r/r_0},$$

and therefore:

$$v^2(r) \propto r\left[\frac{dm}{dr} + \frac{R_0}{r_0}e^{-r/r_0}\right].$$

This expression yields:

- steep inner rise (from $dm/dr$),
- transition shoulder (from comparable $dm/dr$ and $dR/dr$),
- flat outer curve (from slowly decaying $dR/dr$).

### *C. Physical Interpretation of the Tension Coherence Scale* $\mathbf{r_0}$

The parameter $r_0$ is inherited from TEEM-CMB residual structures. It corresponds to the coherence length of the tension field after the partial-reset:

$$r_0 \sim \xi_R,$$

where $\xi_R$ is the correlation length of post-reset R/m fluctuations.

This explains:

- dwarf galaxies → small $\xi_R$ → small $r_0$ → large cores
- massive spirals → large $\xi_R$ → large $r_0$ → extended flat curves
- LSB galaxies → intermediate $\xi_R$

Thus, $r_0$ is not a free parameter but a cosmological relic.

### *D. Relation Between TEEM-DM and MOND-like Acceleration*

In the outer regions:

- $v$ is small,
- $R$ is large,
- therefore $v/R \ll 1$.

The slow-time regime implies:

$$a_{\text{eff}} \propto \frac{v}{R}\, a_{\text{Newton}},$$

which mimics MOND-like low-acceleration behavior without modifying Newtonian dynamics. This explains why TEEM reproduces MOND phenomenology naturally.

### *E. Lensing Potential in TEEM*

The lensing potential is:

$$\Phi_{\text{lens}} \propto m + R.$$

Because $R$ is smooth and coherent:

- small-scale lensing power is reduced,
- subhalo signatures are suppressed,
- convergence maps are smoother than in $\Lambda$CDM.

This provides a clear observational discriminator.

### *F. Summary of Key Parameters*

| Parameter | Meaning | Origin |
|---|---|---|
| $R_0$ | Asymptotic tension amplitude | Residual R-energy from TEEM-CMB |
| $r_0$ | Tension coherence scale | Correlation length of R/m structures |
| $\kappa$ | R-diffusion coefficient | TEEM field dynamics |
| $\lambda$ | m–R coupling | Energetic exchange rules |
| $\eta$ | HF suppression | Partial-reset mechanism |

These parameters are not arbitrary—they arise from TEEM's energetic structure.

### *G. Continuity Across TEEM Frameworks*

| Scale | TEEM Branch | Role of $R$ |
|---|---|---|
| Early Universe | TEEM-CMB | Residual R/m structures formed |
| Galaxies | TEEM-DM | R-gradients shape rotation curves |
| Black Holes | TEEM-BH | R-critical surfaces, horizons, cores |

This demonstrates the unified nature of TEEM across all gravitational regimes.

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